Will the AI build-out pay for itself?

The one-page version of *The Macroeconomic Limits of AI Compute*, version 28 (30 October 2026)

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The question. This year Microsoft, Alphabet, Amazon, Meta and Oracle will spend about $780 billion on AI data centers. Musk's SpaceX/xAI, CoreWeave, Nebius and others add about $140 billion more. Will the money come back?

How we count. A Fermi estimate: the answer is built from many small factors, each bounded and sourced, and the uncertainty in each is carried to the result. All 53 inputs sit in a public register; change one, run one script, and the paper rebuilds itself.

The answer in five lines

- **The cost.** The buildings and chips cost about $890 billion a year to own by 2030 — wear, running costs and a return on the money — even if spending stays flat.
- **The revenue.** AI revenue is about $179 billion this year. To cover the cost by 2030 it must grow 57% a year, every year. Counted sector by sector it grows about 33%.
- **The gap.** About $420 billion a year short in 2030. Flat spending is the optimistic case: every company pointed 2027 spending up.
- **The wait.** Counting the early losses and the cost of the money, no spending path earns an ordinary return by 2035. Even a one-third cut turns a yearly profit only from 2032.
- **What would prove it wrong.** AI revenue above $440 billion in calendar 2028.

New in this version. **Every builder counted.** The rent that SpaceX/xAI, CoreWeave and the others earn from AI labs and clouds was always in the revenue we counted; their buildings and chips were not in the cost. Now they are. That one correction moves the 2030 gap from about $260 billion (the five largest alone) to about $420 billion. Builders that publish no figure yet — Meta's off-balance-sheet Hyperion site, Anthropic's own sites, Crusoe, Lambda, Nscale, Tesla, the Gulf's Humain and Stargate UAE — are listed with what will bring each in. Every one of them would widen the gap.

Also new. *Inflation*, counted once: after it, flat spending is losing about 6.5% a year in 2035. *Tax*: the new write-off delays tax rather than cancelling it, so tax widens the gap. *The October earnings*: Alphabet lost about $500 billion of value in a day on its spending; Anthropic's first filed numbers matched the paper's estimate.

Why it slows. Time saved is not money saved; checking AI's work takes time; much of the value goes to buyers and integrators, not the cloud; countries build their own clouds; and the grid and local consent hold projects back.

What it means. Not a crash, and not the end of AI. The likely path is write-downs and a spending retreat in 2027–29, then a cheaper, more distributed build. The question for investors is not whether AI is useful. It is how long they wait, and what the wait costs.

Where I was wrong. Earlier versions counted the other builders' rent but not their cost, and printed one edge-computing figure a thousand times too large. Both are corrected and listed, as every error will be.

Read more.

Paper, version 28: https://doi.org/[v28DOI] All versions: https://doi.org/10.5281/zenodo.23406475

Deck (19 slides): on the Zenodo record Model, inputs and code: https://github.com/desfitz/ai-compute-limits

What changed: https://github.com/desfitz/ai-compute-limits/blob/main/CHANGES.md First open question: https://github.com/desfitz/ai-compute-limits/issues/1

SSRN: https://papers.ssrn.com/sol3/papers.cfm?abstract_id=7620359 Data Democracy: https://datademocracy.substack.com/p/the-macroeconomic-limits-of-ai-compute

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